

PROJECT REPORT

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Student Performance Predictor using Machine Learning

1. Introduction

Artificial Intelligence (AI) and Machine Learning (ML) are transforming many fields, including education. Machine Learning allows systems to learn from data and make predictions without being explicitly programmed. In the education sector, ML can be used to analyze student behavior and predict academic performance.

This project focuses on building a simple machine learning model that predicts whether a student will pass or fail based on basic factors such as study hours, attendance, and sleep patterns.

2. Problem Statement

Many students are unable to assess their academic performance in advance. They often realize their weaknesses only after receiving exam results. There is a need for a system that can predict student performance early so that necessary improvements can be made.

3. Objective

The main objectives of this project are:

- To develop a machine learning model that predicts student performance
- To understand the basics of supervised learning

- To apply classification techniques using simple data
- To demonstrate how AI can be used in the education domain

4. Methodology

The project follows these steps:

1. Data Collection

A small dataset is manually created containing:

- Study hours
- Attendance percentage
- Sleep hours
- Result (Pass/Fail)

2. Data Preparation

The dataset is converted into a structured format using a data processing library.

3. Model Selection

A **Logistic Regression** algorithm is used because:

- It is simple and easy to understand
- Suitable for classification problems

4. Training the Model

The dataset is divided into:

- Training data
- Testing data

The model learns patterns from the training data.

5. Prediction

The trained model is used to predict the result for new input values.

5. Code Explanation

- The dataset is created using Python dictionary and converted into a DataFrame
- Input features (study hours, attendance, sleep hours) are stored in variable X
- Output (result) is stored in variable y
- The dataset is split into training and testing sets
- A Logistic Regression model is created and trained
- New input data is given to the model for prediction
- The model outputs either:
 - 0 (Fail)
 - 1 (Pass)

6. Result

The model successfully predicts student performance based on the given inputs. For example:

- Input: Study Hours = 4, Attendance = 70, Sleep = 7
- Output: **1 (Pass)**

This shows that the model is able to learn patterns and make predictions.

7. Conclusion

This project demonstrates how Machine Learning can be used to solve real-world problems in education. Even with a small dataset, the model is able to provide useful predictions.

It also helps in understanding the fundamentals of AI and ML, including data handling, model training, and prediction.

8. Future Scope

- Use a larger and real dataset
- Improve model accuracy
- Add more features (like assignment scores, test marks)
- Develop a graphical user interface (GUI)
- Convert the project into a web application

9. References

- Python Documentation
- Scikit-learn Documentation
- Online tutorials and learning resources